

Task H - Feature Scaling Decision Report (Page 1 of 12)

| Feature        | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|----------------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| isFraud        | Standardization | High skewness (5.06); Standardization is robust against extreme varia |  | 0.035                | 0.1838              |
| TransactionDT  | Standardization | Continuous numerical feature; Standardization selected to center data |  | 7372291.1588         | 4617221.098         |
| TransactionAmt | Standardization | High skewness (14.37); Standardization is robust against extreme var  |  | 135.0268             | 239.1626            |
| card1          | Standardization | Continuous numerical feature; Standardization selected to center data |  | 9898.7646            | 4901.1605           |
| card2          | Standardization | Continuous numerical feature; Standardization selected to center data |  | 362.5556             | 156.5954            |
| card3          | Standardization | High skewness (2.02); Standardization is robust against extreme varia |  | 153.1865             | 11.3226             |
| card5          | Standardization | Continuous numerical feature; Standardization selected to center data |  | 199.4716             | 41.1577             |
| card6          | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.7477               | 0.4343              |
| addr1          | Standardization | Continuous numerical feature; Standardization selected to center data |  | 290.7341             | 95.9142             |
| addr2          | Standardization | High skewness (-15.39); Standardization is robust against extreme va  |  | 86.8228              | 2.5373              |
| dist1          | Standardization | High skewness (8.23); Standardization is robust against extreme varia |  | 52.5839              | 242.3519            |
| P_emaildomain  | Standardization | High skewness (3.91); Standardization is robust against extreme varia |  | 1.8335               | 3.7573              |
| C1             | Standardization | High skewness (23.96); Standardization is robust against extreme var  |  | 14.0923              | 133.5693            |
| C2             | Standardization | High skewness (23.68); Standardization is robust against extreme var  |  | 15.2696              | 154.6692            |
| C3             | Standardization | High skewness (88.95); Standardization is robust against extreme var  |  | 0.0056               | 0.1505              |
| C4             | Standardization | High skewness (22.08); Standardization is robust against extreme var  |  | 4.0922               | 68.8486             |
| C5             | Standardization | High skewness (5.79); Standardization is robust against extreme varia |  | 5.5713               | 25.7866             |
| C6             | Standardization | High skewness (19.77); Standardization is robust against extreme var  |  | 9.071                | 71.5086             |
| C7             | Standardization | High skewness (27.19); Standardization is robust against extreme var  |  | 2.8485               | 61.7275             |
| C8             | Standardization | High skewness (26.08); Standardization is robust against extreme var  |  | 5.1446               | 95.3788             |

Task H - Feature Scaling Decision Report (Page 2 of 12)

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| C9      | Standardization | High skewness (5.69); Standardization is robust against extreme varia |  | 4.4801               | 16.6746             |
| C10     | Standardization | High skewness (25.22); Standardization is robust against extreme var  |  | 5.2404               | 95.5817             |
| C11     | Standardization | High skewness (22.36); Standardization is robust against extreme var  |  | 10.2414              | 94.3365             |
| C12     | Standardization | High skewness (27.42); Standardization is robust against extreme var  |  | 4.0762               | 86.6664             |
| C13     | Standardization | High skewness (8.99); Standardization is robust against extreme varia |  | 32.5393              | 129.364             |
| C14     | Standardization | High skewness (16.54); Standardization is robust against extreme var  |  | 8.2951               | 49.5442             |
| D1      | Standardization | Continuous numerical feature; Standardization selected to center data |  | 94.1512              | 157.5481            |
| D2      | Standardization | Continuous numerical feature; Standardization selected to center data |  | 135.0602             | 133.4326            |
| D3      | Standardization | High skewness (6.23); Standardization is robust against extreme varia |  | 19.2875              | 47.5565             |
| D4      | Standardization | Continuous numerical feature; Standardization selected to center data |  | 107.3927             | 169.4888            |
| D5      | Standardization | High skewness (5.20); Standardization is robust against extreme varia |  | 25.3701              | 63.4496             |
| D10     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 109.9524             | 174.3201            |
| D11     | Standardization | High skewness (2.06); Standardization is robust against extreme varia |  | 97.6152              | 144.6349            |
| D15     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 163.7445             | 186.806             |
| M1      | Standardization | High skewness (153.68); Standardization is robust against extreme va  |  | 0.0                  | 0.0065              |
| M2      | Standardization | High skewness (3.80); Standardization is robust against extreme varia |  | 0.0575               | 0.2328              |
| M3      | Standardization | High skewness (2.42); Standardization is robust against extreme varia |  | 0.1147               | 0.3186              |
| M5      | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.1821               | 0.386               |
| M6      | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.6726               | 0.4693              |
| M7      | Standardization | High skewness (3.87); Standardization is robust against extreme varia |  | 0.0557               | 0.2294              |

Task H - Feature Scaling Decision Report (Page 3 of 12)

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| M8      | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.1508               | 0.3578              |
| M9      | Standardization | High skewness (3.52); Standardization is robust against extreme varia |  | 0.0654               | 0.2473              |
| V1      | Standardization | High skewness (-186.37); Standardization is robust against extreme v  |  | 1.0                  | 0.0054              |
| V2      | Standardization | High skewness (9.44); Standardization is robust against extreme varia |  | 1.0238               | 0.1758              |
| V3      | Standardization | High skewness (7.56); Standardization is robust against extreme varia |  | 1.0412               | 0.2362              |
| V4      | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.8465               | 0.3195              |
| V5      | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.877                | 0.3455              |
| V6      | Standardization | High skewness (9.37); Standardization is robust against extreme varia |  | 1.0241               | 0.1753              |
| V7      | Standardization | High skewness (7.73); Standardization is robust against extreme varia |  | 1.0384               | 0.2242              |
| V8      | Standardization | High skewness (11.94); Standardization is robust against extreme var  |  | 1.0146               | 0.1358              |
| V9      | Standardization | High skewness (9.67); Standardization is robust against extreme varia |  | 1.0219               | 0.1656              |
| V10     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.4639               | 0.3786              |
| V11     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.479                | 0.4011              |
| V12     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.5597               | 0.4765              |
| V13     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.5992               | 0.4967              |
| V14     | Standardization | High skewness (-47.90); Standardization is robust against extreme va  |  | 0.9996               | 0.0209              |
| V15     | Standardization | High skewness (2.84); Standardization is robust against extreme varia |  | 0.1066               | 0.313               |
| V16     | Standardization | High skewness (4.15); Standardization is robust against extreme varia |  | 0.1076               | 0.3227              |
| V17     | Standardization | High skewness (4.10); Standardization is robust against extreme varia |  | 0.1168               | 0.3431              |
| V18     | Standardization | High skewness (4.33); Standardization is robust against extreme varia |  | 0.1179               | 0.3499              |

Task H - Feature Scaling Decision Report (Page 4 of 12)

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| V19     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.8164               | 0.3972              |
| V20     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.8478               | 0.4288              |
| V21     | Standardization | High skewness (2.55); Standardization is robust against extreme varia |  | 0.113                | 0.3194              |
| V22     | Standardization | High skewness (3.94); Standardization is robust against extreme varia |  | 0.1153               | 0.3384              |
| V23     | Standardization | High skewness (12.99); Standardization is robust against extreme var  |  | 1.0303               | 0.2315              |
| V24     | Standardization | High skewness (9.74); Standardization is robust against extreme varia |  | 1.0506               | 0.2858              |
| V25     | Standardization | High skewness (-2.88); Standardization is robust against extreme vari |  | 0.9805               | 0.1731              |
| V26     | Standardization | High skewness (2.85); Standardization is robust against extreme varia |  | 0.9896               | 0.1954              |
| V27     | Standardization | High skewness (43.47); Standardization is robust against extreme var  |  | 0.0007               | 0.0267              |
| V28     | Standardization | High skewness (45.95); Standardization is robust against extreme var  |  | 0.0007               | 0.0291              |
| V29     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.3878               | 0.4766              |
| V30     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.3541               | 0.535               |
| V31     | Standardization | High skewness (2.69); Standardization is robust against extreme varia |  | 0.1226               | 0.3354              |
| V32     | Standardization | High skewness (3.83); Standardization is robust against extreme varia |  | 0.1241               | 0.347               |
| V33     | Standardization | High skewness (2.58); Standardization is robust against extreme varia |  | 0.1139               | 0.3212              |
| V34     | Standardization | High skewness (3.22); Standardization is robust against extreme varia |  | 0.1212               | 0.337               |
| V35     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.5426               | 0.436               |
| V36     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.5792               | 0.4554              |
| V37     | Standardization | High skewness (26.66); Standardization is robust against extreme var  |  | 1.0771               | 0.5855              |
| V38     | Standardization | High skewness (21.59); Standardization is robust against extreme var  |  | 1.1159               | 0.7294              |

Task H - Feature Scaling Decision Report (Page 5 of 12)

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| V39     | Standardization | High skewness (7.00); Standardization is robust against extreme varia |  | 0.1186               | 0.3892              |
| V40     | Standardization | High skewness (7.80); Standardization is robust against extreme varia |  | 0.1265               | 0.4348              |
| V41     | Standardization | High skewness (-43.75); Standardization is robust against extreme va  |  | 0.9995               | 0.0228              |
| V42     | Standardization | High skewness (3.07); Standardization is robust against extreme varia |  | 0.1114               | 0.3311              |
| V43     | Standardization | High skewness (3.80); Standardization is robust against extreme varia |  | 0.1206               | 0.374               |
| V44     | Standardization | High skewness (34.25); Standardization is robust against extreme var  |  | 1.0599               | 0.5413              |
| V45     | Standardization | High skewness (27.33); Standardization is robust against extreme var  |  | 1.0862               | 0.619               |
| V46     | Standardization | High skewness (9.25); Standardization is robust against extreme varia |  | 1.0159               | 0.1412              |
| V47     | Standardization | High skewness (11.91); Standardization is robust against extreme var  |  | 1.0275               | 0.1967              |
| V48     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.3832               | 0.4294              |
| V49     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.2839               | 0.4925              |
| V50     | Standardization | High skewness (2.49); Standardization is robust against extreme varia |  | 0.1176               | 0.3246              |
| V51     | Standardization | High skewness (3.04); Standardization is robust against extreme varia |  | 0.1218               | 0.3499              |
| V52     | Standardization | High skewness (4.01); Standardization is robust against extreme varia |  | 0.1304               | 0.38                |
| V53     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.5776               | 0.477               |
| V54     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.62                 | 0.4985              |
| V55     | Standardization | High skewness (11.18); Standardization is robust against extreme var  |  | 1.0588               | 0.3656              |
| V56     | Standardization | High skewness (25.17); Standardization is robust against extreme var  |  | 1.1052               | 0.6178              |
| V57     | Standardization | High skewness (2.94); Standardization is robust against extreme varia |  | 0.1116               | 0.3284              |
| V58     | Standardization | High skewness (4.11); Standardization is robust against extreme varia |  | 0.1152               | 0.3506              |

Task H - Feature Scaling Decision Report (Page 6 of 12)

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| V59     | Standardization | High skewness (4.72); Standardization is robust against extreme varia |  | 0.1169               | 0.3566              |
| V60     | Standardization | High skewness (5.37); Standardization is robust against extreme varia |  | 0.1239               | 0.3929              |
| V61     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.8298               | 0.4071              |
| V62     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.8676               | 0.4511              |
| V63     | Standardization | High skewness (3.10); Standardization is robust against extreme varia |  | 0.1137               | 0.3342              |
| V64     | Standardization | High skewness (4.17); Standardization is robust against extreme varia |  | 0.1233               | 0.3817              |
| V65     | Standardization | High skewness (-58.40); Standardization is robust against extreme va  |  | 0.9997               | 0.0171              |
| V66     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.981                | 0.2016              |
| V67     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.9984               | 0.2293              |
| V68     | Standardization | High skewness (50.08); Standardization is robust against extreme var  |  | 0.0005               | 0.0222              |
| V69     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.3902               | 0.479               |
| V70     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.3547               | 0.535               |
| V71     | Standardization | High skewness (2.82); Standardization is robust against extreme varia |  | 0.1223               | 0.343               |
| V72     | Standardization | High skewness (3.92); Standardization is robust against extreme varia |  | 0.1262               | 0.3662              |
| V73     | Standardization | High skewness (2.91); Standardization is robust against extreme varia |  | 0.1217               | 0.3455              |
| V74     | Standardization | High skewness (3.08); Standardization is robust against extreme varia |  | 0.1323               | 0.3703              |
| V75     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.5443               | 0.4739              |
| V76     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.5876               | 0.4959              |
| V77     | Standardization | High skewness (14.35); Standardization is robust against extreme var  |  | 1.0738               | 0.4921              |
| V78     | Standardization | High skewness (16.53); Standardization is robust against extreme var  |  | 1.1227               | 0.7219              |

Task H - Feature Scaling Decision Report (Page 7 of 12)

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| V79     | Standardization | High skewness (4.26); Standardization is robust against extreme varia |  | 0.1162               | 0.3535              |
| V80     | Standardization | High skewness (6.17); Standardization is robust against extreme varia |  | 0.1222               | 0.3812              |
| V81     | Standardization | High skewness (6.44); Standardization is robust against extreme varia |  | 0.1296               | 0.4203              |
| V82     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.8446               | 0.3893              |
| V83     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.882                | 0.4338              |
| V84     | Standardization | High skewness (3.03); Standardization is robust against extreme varia |  | 0.1164               | 0.337               |
| V85     | Standardization | High skewness (4.11); Standardization is robust against extreme varia |  | 0.1272               | 0.3902              |
| V86     | Standardization | High skewness (18.35); Standardization is robust against extreme var  |  | 1.0551               | 0.3875              |
| V87     | Standardization | High skewness (16.02); Standardization is robust against extreme var  |  | 1.0844               | 0.4724              |
| V88     | Standardization | High skewness (-39.49); Standardization is robust against extreme va  |  | 0.9994               | 0.0253              |
| V89     | Standardization | High skewness (41.89); Standardization is robust against extreme var  |  | 0.0008               | 0.0294              |
| V90     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.4019               | 0.4756              |
| V91     | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.357                | 0.5392              |
| V92     | Standardization | High skewness (2.83); Standardization is robust against extreme varia |  | 0.1276               | 0.3503              |
| V93     | Standardization | High skewness (3.94); Standardization is robust against extreme varia |  | 0.1314               | 0.3748              |
| V94     | Standardization | High skewness (2.39); Standardization is robust against extreme varia |  | 0.1163               | 0.3206              |
| V95     | Standardization | High skewness (30.25); Standardization is robust against extreme var  |  | 1.0375               | 21.0288             |
| V96     | Standardization | High skewness (26.04); Standardization is robust against extreme var  |  | 3.0036               | 40.2289             |
| V97     | Standardization | High skewness (25.33); Standardization is robust against extreme var  |  | 1.718                | 27.6932             |
| V98     | Standardization | High skewness (6.39); Standardization is robust against extreme varia |  | 0.062                | 0.2849              |

Task H - Feature Scaling Decision Report (Page 8 of 12)

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| V99     | Standardization | High skewness (11.07); Standardization is robust against extreme var  |  | 0.8945               | 2.7219              |
| V100    | Standardization | High skewness (8.74); Standardization is robust against extreme varia |  | 0.2734               | 0.9469              |
| V101    | Standardization | High skewness (30.64); Standardization is robust against extreme var  |  | 0.8888               | 20.5772             |
| V102    | Standardization | High skewness (27.00); Standardization is robust against extreme var  |  | 1.8263               | 35.919              |
| V103    | Standardization | High skewness (26.04); Standardization is robust against extreme var  |  | 1.2786               | 25.6789             |
| V104    | Standardization | High skewness (14.89); Standardization is robust against extreme var  |  | 0.0854               | 0.6485              |
| V105    | Standardization | High skewness (20.90); Standardization is robust against extreme var  |  | 0.281                | 3.3731              |
| V106    | Standardization | High skewness (19.49); Standardization is robust against extreme var  |  | 0.1645               | 1.8252              |
| V107    | Standardization | High skewness (-48.77); Standardization is robust against extreme va  |  | 0.9996               | 0.0205              |
| V108    | Standardization | High skewness (17.83); Standardization is robust against extreme var  |  | 1.0046               | 0.0811              |
| V109    | Standardization | High skewness (9.73); Standardization is robust against extreme varia |  | 1.0148               | 0.1277              |
| V110    | Standardization | High skewness (13.75); Standardization is robust against extreme var  |  | 1.0077               | 0.0973              |
| V111    | Standardization | High skewness (31.72); Standardization is robust against extreme var  |  | 1.0026               | 0.0708              |
| V112    | Standardization | High skewness (23.15); Standardization is robust against extreme var  |  | 1.0054               | 0.0846              |
| V113    | Standardization | High skewness (28.97); Standardization is robust against extreme var  |  | 1.0034               | 0.0747              |
| V114    | Standardization | High skewness (13.06); Standardization is robust against extreme var  |  | 1.0093               | 0.1102              |
| V115    | Standardization | High skewness (6.59); Standardization is robust against extreme varia |  | 1.0324               | 0.1903              |
| V116    | Standardization | High skewness (9.79); Standardization is robust against extreme varia |  | 1.0157               | 0.1367              |
| V117    | Standardization | High skewness (9.38); Standardization is robust against extreme varia |  | 1.0004               | 0.0352              |
| V118    | Standardization | High skewness (21.56); Standardization is robust against extreme var  |  | 1.0015               | 0.041               |

Task H - Feature Scaling Decision Report (Page 9 of 12)

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| V119    | Standardization | High skewness (15.48); Standardization is robust against extreme var  |  | 1.0007               | 0.0364              |
| V120    | Standardization | High skewness (12.85); Standardization is robust against extreme var  |  | 1.0009               | 0.0417              |
| V121    | Standardization | High skewness (14.37); Standardization is robust against extreme var  |  | 1.0043               | 0.0671              |
| V122    | Standardization | High skewness (15.71); Standardization is robust against extreme var  |  | 1.0018               | 0.0486              |
| V123    | Standardization | High skewness (12.88); Standardization is robust against extreme var  |  | 1.0311               | 0.2281              |
| V124    | Standardization | High skewness (6.09); Standardization is robust against extreme varia |  | 1.0929               | 0.374               |
| V125    | Standardization | High skewness (9.00); Standardization is robust against extreme varia |  | 1.0504               | 0.28                |
| V126    | Standardization | High skewness (29.59); Standardization is robust against extreme var  |  | 129.9107             | 2346.3355           |
| V127    | Standardization | High skewness (25.69); Standardization is robust against extreme var  |  | 336.4312             | 4237.5574           |
| V128    | Standardization | High skewness (25.06); Standardization is robust against extreme var  |  | 203.9858             | 3009.4696           |
| V129    | Standardization | High skewness (240.34); Standardization is robust against extreme va  |  | 8.7643               | 113.803             |
| V130    | Standardization | High skewness (26.30); Standardization is robust against extreme var  |  | 92.1161              | 315.8837            |
| V131    | Standardization | High skewness (89.62); Standardization is robust against extreme var  |  | 31.1167              | 161.1204            |
| V132    | Standardization | High skewness (30.24); Standardization is robust against extreme var  |  | 103.4584             | 2265.5106           |
| V133    | Standardization | High skewness (26.87); Standardization is robust against extreme var  |  | 204.7804             | 3795.3199           |
| V134    | Standardization | High skewness (25.99); Standardization is robust against extreme var  |  | 145.8949             | 2772.2586           |
| V135    | Standardization | High skewness (144.92); Standardization is robust against extreme va  |  | 17.241               | 293.7704            |
| V136    | Standardization | High skewness (49.03); Standardization is robust against extreme var  |  | 38.7998              | 451.6897            |
| V137    | Standardization | High skewness (90.61); Standardization is robust against extreme var  |  | 26.3512              | 348.2415            |
| V279    | Standardization | High skewness (30.24); Standardization is robust against extreme var  |  | 1.123                | 21.0218             |

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| V280    | Standardization | High skewness (24.78); Standardization is robust against extreme var  |  | 1.967                | 27.8516             |
| V281    | Standardization | High skewness (10.02); Standardization is robust against extreme var  |  | 0.0876               | 0.5122              |
| V282    | Standardization | High skewness (3.83); Standardization is robust against extreme varia |  | 0.8176               | 0.9209              |
| V283    | Standardization | High skewness (12.41); Standardization is robust against extreme var  |  | 0.9911               | 1.5571              |
| V284    | Standardization | High skewness (5.30); Standardization is robust against extreme varia |  | 0.0885               | 0.3384              |
| V285    | Standardization | High skewness (12.29); Standardization is robust against extreme var  |  | 1.1676               | 3.2824              |
| V286    | Standardization | High skewness (7.59); Standardization is robust against extreme varia |  | 0.0315               | 0.1909              |
| V287    | Standardization | High skewness (8.84); Standardization is robust against extreme varia |  | 0.3586               | 1.079               |
| V288    | Standardization | High skewness (2.56); Standardization is robust against extreme varia |  | 0.184                | 0.4306              |
| V289    | Standardization | High skewness (3.51); Standardization is robust against extreme varia |  | 0.2355               | 0.5987              |
| V290    | Standardization | High skewness (34.57); Standardization is robust against extreme var  |  | 1.103                | 0.7689              |
| V291    | Standardization | High skewness (50.16); Standardization is robust against extreme var  |  | 1.6598               | 16.2524             |
| V292    | Standardization | High skewness (51.23); Standardization is robust against extreme var  |  | 1.2399               | 3.775               |
| V293    | Standardization | High skewness (30.58); Standardization is robust against extreme var  |  | 0.9426               | 20.5887             |
| V294    | Standardization | High skewness (23.85); Standardization is robust against extreme var  |  | 2.3138               | 39.5262             |
| V295    | Standardization | High skewness (25.34); Standardization is robust against extreme var  |  | 1.4334               | 25.9627             |
| V296    | Standardization | High skewness (18.95); Standardization is robust against extreme var  |  | 0.3282               | 3.2613              |
| V297    | Standardization | High skewness (13.28); Standardization is robust against extreme var  |  | 0.089                | 0.6283              |
| V298    | Standardization | High skewness (19.89); Standardization is robust against extreme var  |  | 0.2988               | 3.175               |
| V299    | Standardization | High skewness (18.68); Standardization is robust against extreme var  |  | 0.1717               | 1.7242              |

Task H - Feature Scaling Decision Report (Page 11 of 12)

| Feature | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|---------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| V300    | Standardization | High skewness (10.86); Standardization is robust against extreme var  |  | 0.0454               | 0.2893              |
| V301    | Standardization | High skewness (10.55); Standardization is robust against extreme var  |  | 0.0519               | 0.318               |
| V302    | Standardization | High skewness (2.80); Standardization is robust against extreme varia |  | 0.2518               | 0.4819              |
| V303    | Standardization | High skewness (5.16); Standardization is robust against extreme varia |  | 0.2831               | 0.6236              |
| V304    | Standardization | High skewness (3.29); Standardization is robust against extreme varia |  | 0.2642               | 0.5282              |
| V305    | Standardization | High skewness (384.23); Standardization is robust against extreme va  |  | 1.0                  | 0.0026              |
| V306    | Standardization | High skewness (29.27); Standardization is robust against extreme var  |  | 139.7463             | 2348.8318           |
| V307    | Standardization | High skewness (23.67); Standardization is robust against extreme var  |  | 408.6725             | 4391.9593           |
| V308    | Standardization | High skewness (24.53); Standardization is robust against extreme var  |  | 230.409              | 3021.9014           |
| V309    | Standardization | High skewness (224.88); Standardization is robust against extreme va  |  | 10.9958              | 116.2534            |
| V310    | Standardization | High skewness (19.50); Standardization is robust against extreme var  |  | 118.1924             | 352.9797            |
| V311    | Standardization | High skewness (323.83); Standardization is robust against extreme va  |  | 4.2021               | 102.3742            |
| V312    | Standardization | High skewness (73.60); Standardization is robust against extreme var  |  | 39.1731              | 172.1271            |
| V313    | Standardization | High skewness (14.04); Standardization is robust against extreme var  |  | 21.3057              | 95.8052             |
| V314    | Standardization | High skewness (10.76); Standardization is robust against extreme var  |  | 43.2263              | 173.4444            |
| V315    | Standardization | High skewness (12.13); Standardization is robust against extreme var  |  | 26.7495              | 116.7345            |
| V316    | Standardization | High skewness (30.10); Standardization is robust against extreme var  |  | 109.8166             | 2270.0159           |
| V317    | Standardization | High skewness (24.60); Standardization is robust against extreme var  |  | 247.6021             | 3980.0126           |
| V318    | Standardization | High skewness (25.48); Standardization is robust against extreme var  |  | 162.1504             | 2793.3224           |
| V319    | Standardization | High skewness (181.84); Standardization is robust against extreme va  |  | 18.3722              | 332.3023            |

Task H - Feature Scaling Decision Report (Page 12 of 12)

| Feature                | Chosen Method   | Reasoning                                                             |  | Param 1 (Min / Mean) | Param 2 (Max / Std) |
|------------------------|-----------------|-----------------------------------------------------------------------|--|----------------------|---------------------|
| V320                   | Standardization | High skewness (70.52); Standardization is robust against extreme var  |  | 42.0712              | 473.4948            |
| V321                   | Standardization | High skewness (123.56); Standardization is robust against extreme va  |  | 28.3262              | 382.0503            |
| ProductCD_W            | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.7445               | 0.4361              |
| ProductCD_H            | Standardization | High skewness (3.87); Standardization is robust against extreme varia |  | 0.0559               | 0.2298              |
| ProductCD_C            | Standardization | High skewness (2.40); Standardization is robust against extreme varia |  | 0.116                | 0.3203              |
| ProductCD_S            | Standardization | High skewness (6.91); Standardization is robust against extreme varia |  | 0.0197               | 0.1389              |
| ProductCD_R            | Standardization | High skewness (3.57); Standardization is robust against extreme varia |  | 0.0638               | 0.2445              |
| card4_discover         | Standardization | High skewness (9.26); Standardization is robust against extreme varia |  | 0.0113               | 0.1055              |
| card4_mastercard       | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.3204               | 0.4666              |
| card4_visa             | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.6542               | 0.4756              |
| card4_american express | Standardization | High skewness (8.24); Standardization is robust against extreme varia |  | 0.0141               | 0.1179              |
| M4_M2                  | Standardization | High skewness (2.64); Standardization is robust against extreme varia |  | 0.1014               | 0.3018              |
| M4_M0                  | Standardization | Continuous numerical feature; Standardization selected to center data |  | 0.8092               | 0.393               |
| M4_M1                  | Standardization | High skewness (2.88); Standardization is robust against extreme varia |  | 0.0895               | 0.2854              |